

Graphics by PRASHANT CHAUDHARY

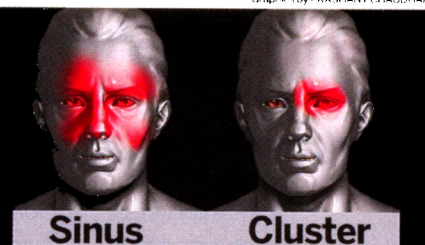
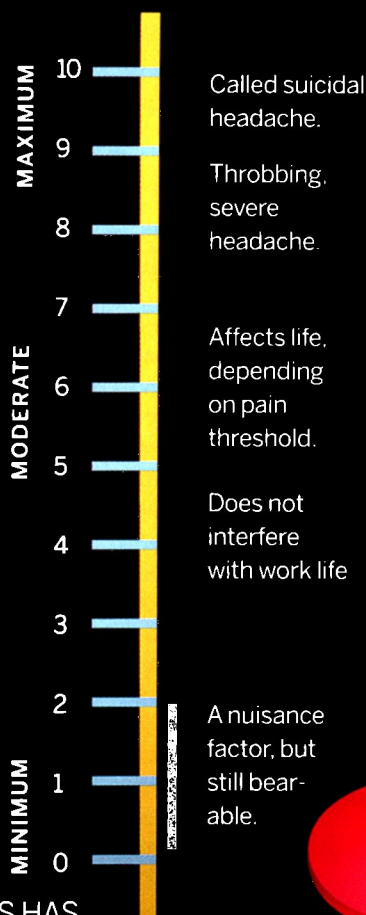
SIGNAL: Nerves release proteins that awaken pain receptors and cause blood vessels on the outer covering of the brain to swell up.

GENERATOR: Signals are sent to trigeminal nerves, a massive network that covers the brain like a helmet

PROCESSOR: Triggers reach hypothalamus, brain's major processing hub responsible for hormones, sleep and hunger.

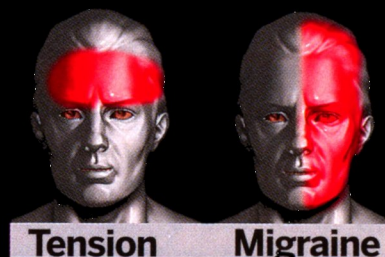
TRIGGERS: Hormonal change, stress, lack of sleep, diet, weather, allergens provide the spark to start the process.

THE BIOLOGY OF HEADACHES HAS LED TO NEW TREATMENTS GEARED TOWARDS CALMING DOWN OVERLY SENSITIVE NERVE CELLS



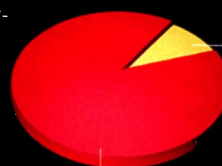
Pain is behind browbone and/or cheekbones.

Pain is in and around one eye.



Pain is like a band squeezing the head

Pain, nausea and visual changes are typical of classic form



Primary headaches
No underlying medical cause

Secondary headaches
Underlying medical causes like tumours, meningitis, vascular disorders, etc.

Source: WHO, 2004; NEJM 2002; XI Congress of the IHS, 2004; Cephalalgia, 2006; Jaslok & Lilavati Hospitals, 2003-06; AIIMS, 2004-2007

required to get rid of one," points out Dr R. Srinivasa, head of neurology at the M.S. Ramaiah Medical College and Hospital in Bangalore. "For a secondary headache, an antibiotic might do the trick." Not so for the more deadly, albeit more uncommon, cluster headache. Typically, it repeats itself, often daily. The pain is so searing that they are also known as 'suicide headaches'. Immediate treatment with oxygen and migraine drugs given intravenously can sometimes provide relief. "The pain from a migraine is a throbbing one, restricted to one side of the head, gets worse with movement and lasts for hours and days." They are ac-

companied by either nausea and vomiting, as they were for Raj, or extreme sensitivity to both light and sound.

At one time, it was believed that headaches are caused by a rapid increase in the flow of blood to the brain. But with new-generation pet scanners, scientists today can tell much more about how the brain works. And it seems the order of events in extreme headaches is not as straightforward as was imagined: "The brain of a sufferer is primed to overreact to all sorts of stimuli that most people can easily tolerate—stress, hormones, food, drink, sleep disturbances," points out Ravishankar. The triggers cause disturbance

of the cerebral cortex, probably the cortical spreading depression, and the wave of excitation stimulates trigeminal nerves (the largest network in the cranium), driving it to release neuropeptides that cause blood vessels to dilate and stretch nerves to maintain a state of alert. The signal finally gets transferred to the pain area of the brain—the thalamus. In brief, swollen blood vessels are the result of a growing migraine, not its cause.

The biology of headaches has led to new treatments, geared towards calming down overly sensitive nerve cells. "Multiple targeted drugs are now available to treat an attack," says Srinivasa.